March 14, 2025

NSF RFI: 90 FR 9088

Organization: Massive Data Institute at the McCourt School of Public Policy, Georgetown University

Subject: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Primary POC: Elissa M. Redmiles, Ph.D., Georgetown University
; this response was compiled by the Primary POC Elissa M. Redmiles, but represents contributions from the collaborators listed below.

In collaboration with:
Lucy Qin, Ph.D., Georgetown University
Allison McDonald, Ph.D., Boston University

The Massive Data Institute (MDI) at the McCourt School of Public Policy at Georgetown University offer the following comments in response to Office of Science and Technology Policy (OSTP) and the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) at the National Science Foundation (NSF)'s *Request for Information on the Development of an Artificial Intelligence (AI) Action Plan.*

The Massive Data Institute at the McCourt School of Public Policy at Georgetown University is an interdisciplinary research institute that connects experts across computer science, data science, public health, public policy, and social science to tackle societal scale issues and impact public policy in a way that improves people's lives through responsible evidence-based research.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Generative AI can offer a superpower [1] to improving the lives of Americans. America's AI action plan can prioritize what Michael Kratsios described in his confirmation hearing as an "American free market approach to scientific discovery" by supporting narrow and specific research into mitigations that protect American families, businesses and national security from specific AI harms like sexualized deepfakes without restricting free-speech or AI progress [2].

American's AI Action plan must protect the sanctity of our families, our national security and our free speech without placing burdensome requirements on private sector AI innovation by:

- ➔ funding scientific research to develop and evaluate mitigations that are narrowly scoped to address sexual deepfakes without infringing on private sector innovation or free speech via NSF sources such as SaTC, TIP, and CAREER and DoD sources such as DARPA, including the YFA program
- ➔ holding technology companies accountable for
  - ◆ implementing narrowly-scoped, privacy- & free-speech preserving mitigations that work to stop the output of sexual deepfakes by generative AI models, the promotion and monetization of tools dedicated to "nudification", and the upload of sexual deepfakes to online platforms
  - ◆ removing child sexual abuse material (CSAM) and non-consensual images depicting adults from training data
- ➔ hosting White House round tables as well as agency listening sessions focused on bringing together technical, legal and social science experts focused on sexual deepfakes to ensure executive orders and regulation centers rigorously evaluated, effective mitigations that will not adversely impact AI progress

Sexual AI Deepfakes are a threat to American Families. The Bipartisan House AI Task Force report [3] states that there have been at least 100,000 victims of sexual deepfakes. The majority of these victims are American children, especially girls. Our research [4] shows that anyone – even those with no technical skills -- can make sexual deepfakes for an average of $0.34 per image.

Sexual AI Deepfakes are a threat to National Security. AI-created deepfakes, particularly sexual ones in which Americans are depicted nude, can be used by foreign actors to blackmail and extort Americans. Prior research finds a majority of those targeted are children, "although cyber sextortion against men by foreign offenders is the most rapidly expanding type of cyber sextortion" [5]. These attacks are coordinated sexual content blackmail schemes in which foreign actors target Americans, eventually getting the target to share sexual images of themselves.

Our ongoing research finds that the proliferation of deepfakes is making it even easier for foreign actors to attack Americans: attackers no longer need the target of sextortion to send sexual images of themselves. As one organization that helps victims recover from sextortion described, attackers “haven't even got to… got to go through the motions of building up that relationship and getting the person to send images anymore, they can just use their profile picture and make an image.”

Tackling sexual deepfakes does not require slowing AI development. Instead, a forward-looking ecosystem-wide agenda of research and policy can ensure American families & national security remain secure while America continues to lead AI progress.

To ensure this future, we must engage in **research to forecast how attackers might use AI** so we can narrowly and specifically define potential harms and develop mitigations. **By engaging in scientific research early we can work in parallel with AI innovation to stop harms before they happen.** Exemplar research directions include technical research to develop computational metrics that can be used to detect when someone’s likeness is being illicitly output by an AI model, model adjustment and post-processing techniques that stop the generation of child sexual abuse, as well as sociotechnical research to develop resilience against deepfakes in the American public or strengthen young computer scientist’s skills in AI threat forecasting and mitigation.

As part of developing and evaluating mitigations against the output of sexual deepfakes from closed-source AI models without slowing the progress of model development, **it is important to evaluate proposed mitigations**. Specifically, how well mitigations will work in practice and what consequences may be AI model performance and free speech. For example, our ongoing research evaluates the proposed mitigation of “concept cleaning” from generative models [6]: to prevent harmful content generation (e.g., child sexual abuse material), concept cleaning proposes the removal or separation of training data related to the harmful output (e.g. images of children, images of nude adults). We attempted 5 different approaches to implementing this proposal. Our initial results [7] suggest that due to labeling errors, concept fuzziness, and model compositionality the mitigation may (a) be less effective than hoped and (b) significantly erode model performance due to needing to remove many related concepts. **Engaging in such evaluations requires significant funding and expertise;** for example the evaluation described here required retraining models from scratch and a team of 5 Ph.D.-trained computer scientists.

Many companies, not just AI model providers, are part of the pipeline of sexual deepfakes. **Engaging in mitigations across the technology ecosystem** will be more effective and less detrimental to AI progress than only focusing on AI models and the companies that create them.

For example, our research studying 20 of the most popular nudification websites, many of which have been used on children [8,9], finds that cryptocurrency is available as a payment method on 17 out of 20 of the websites we studied [4]. **The use of cryptocurrency to process payments for AI sexual deepfakes threatens the legitimacy of cryptocurrency**. Paypal and Visa are advertised as processing payments on 8 and 7 of 20 websites we study, respectively. Cryptocurrency exchanges and mainstream payment processors need to be held accountable for their role in powering the sexual deepfake ecosystem.

Similarly, **social media platforms & app store providers can prevent the promotion of sexual deepfake creation tools**. Reporting from 404 Media found that five different nudification applications were being advertised on Meta's platforms [10]. Platforms must be held accountable for such ads and be urged to proactively identify and remove similar ads. Similarly, apps that offer to create sexual deepfakes of Americans should be banned from app stores without the need for journalists to raise awareness before such action is taken [11].

Finally, research has found that real child sexual abuse material [12] and adult non-consensual intimate images [13] are being used by American AI companies and researchers to train their AI models. **We must ensure that American's AI progress is not powered by sexual abuse material.**

References

[1] IMPAULSIVE. 2024. The Donald Trump Interview EP. 418. Retrieved from https://www.youtube.com/watch?v=xrFdHO7FH8w.

[2] The Heritage Foundation. 2018. Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security. Retrieved from https://www.heritage.org/homeland-security/event/deep-fakes-looming-challenge-privacy-democracy-and-national-security.

[3] House Bipartisan Task Force on Artificial Intelligence. 2024. Bipartisan House Task Force Report on Artificial Intelligence – 118th Congress. Retrieved from https://science.house.gov/2024/12/house-bipartisan-task-force-on-artificial-intelligence-delivers-report.

[4] Gibson, C., Olszewski, D., Brigham, N. G., Crowder, A., Butler, K. R. B., Traynor, P., Redmiles, E. M., and Kohno T.. Forthcoming. Analyzing the AI Nudification Application Ecosystem. To appear in USENIX Security 2025. Retrieved from https://arxiv.org/abs/2411.09751.

[5] O'Malley, R. L., & Holt, K. M. 2022. Cyber sextortion: An exploratory analysis of different perpetrators engaging in a similar crime. Journal of interpersonal violence, 37(1-2), 258-283.

[6] Thorn and All Tech Is Human. 2024. Safety by Design for Generative AI: Preventing Child Sexual Abuse. Retrieved from https://info.thorn.org/hubfs/thorn-safety-by-design-for-generative-AI.pdf.

[7] Redmiles, E.M., Qin, L., Bargal, S.A., Cretu, A., Kireev, K., Troncoso, C., Brigham, N.G., Wei, M., Kohno, T., Cintaqia, P., McDonald, A., Aya, A., Kumar, D., Butler, K., Gibson, C., Tranor, P., Eaton, A., AI can create synthetic non-consensual intimate imagery. Retrieved from: https://georgetown.box.com/v/ai-ibsa-onepager.

[8] U.S. Department of Justice. 2024. Recidivist Sex Offender Sentenced for Possessing Deepfake Child Sexual Abuse Material. Retrieved from: https://www.justice.gov/archives/opa/pr/recidivist-sex-offender-sentenced-possessing-deepfake-child-sexual-abuse-material.

[9] Verma, P. and Harwell, D. 2024. In novel case, U.S. charges man with making child sex abuse images with AI. Retrieved from: http://washingtonpost.com/technology/2024/05/21/doj-arrest-ai-csam-child-sexual-abuse-images/.

[10] Maiberg, E. 2024. Instagram's Nudify Ads. 404 Media. Retrieved from https://www.404media.co/email/d2bebba9-5808-44fc-8352-d93d1791a5ff/?ref=daily-stories-newsletter.

[11] Maiberg, E. 2024. Apple Removes Nonconsensual AI Nude Apps Following 404 Media Investigation. 404 Media. Retrieved from https://www.404media.co/apple-removes-nonconsensual-ai-nude-apps-following-404-media-investigation/.

[12] Thiel, D. 2023. Stanford Internet Observatory, Cyber Policy Center. Identifying and eliminating CSAM in generative ML training data and models. Retrieved from https://stacks.stanford.edu/file/druid:kh752sm9123/ml_training_data_csam_report-2023-12-23.pdf.

[13] Arya, A., Cintaqia, P., Kumar, D., McDonald, A., Qin, L., and Redmiles, E. M.. 2024. NeurIPS EvalEval Workshop. (Mis)use of nude images in machine learning research. Retrieved from https://evaleval.github.io/accepted_papers/EvalEval_24_Arya.pdf.